**Outline for CS Capstone Project Document**

**This capstone project composes of**

Front cover

Inside cover and approval page

Abstract

Acknowledgements

Table of Contents

List of Tables

List of Figures

**Chapter 1 Introduction** (suggestions for the contents of Chapters 1-5 follows below)

1.1 Background information

1.2 Objectives

1.3 Scope

1.4 Expected benefits

**Chapter 2 Feasibility Study**

2.1 Introduction

2.2 Problem statement

2.3 Related research and projects

2.4 Requirement specifications for the new system

2.5 Implementation techniques

2.6 Implementation plan

**Chapter 3 Analysis and Design**

3.1 Introduction

3.2 Analysis of the existing system

3.3 User requirement analysis

3.4 System design

**Chapter 4 System Functionality**

4.1 Introduction

4.2 System Architecture

4.3 Test plan

4.4 Test results

**Chapter 5 Summary and Suggestions**

5.1 Project Summary

5.2 Problems encountered and solutions

5.3 Suggestions for further development

References

Appendix

**Example and suggestions for each chapter's content**

**Chapter 1 Introduction**

1.1 Background

- Short description of the current system
- Who, what, where, when, why

(Sample) Our Institute has a Student Services Division to act as the center for managing student activities and provide various student services. The Student Services Division manages many student activities and services. There are numerous requests for each of those services, hence a large amount of data and information to be managed and processed. The sources of data and information include various units within the Institute as well as other external organizations. Currently there is no system to manage this information making it difficult for the staff to access the information they need and perform their duties. This situation results in delays and frustration for students and others who interact and work with the Student Services Division. Thus it is necessary to implement a database system and applications to manage data and information so as to provide the staff of the SSD with a tool to help them in performing their jobs.

1.2 Objectives

- What system (s) will you build? Why?

(Sample) This project aims to create a software system that uses a database management system to store and retrieve data and information for the Student Services Division. This software should help the staff of SSD to perform their jobs more efficiently and effectively.

1.3 Scope

- Initial scope only, modifications/changes will be made to this in the Requirement Specifications after the feasibility study
- functions and features
- users
- related systems (software and hardware)
- supported platforms

(Sample) This project is a web-based database system with the following features

- users can add, edit, delete data and fields in the database
- The information contained in the database include extracurricular and academic activities that is not recorded by the Registra's Office
- Users can use all the functionality of the system through a web browser
- The system includes the following student activity reports
  (1) Memberships in student organizations and groups
  (2) Attendance of Faculty or University activities
- Users are separated into Administrators and Staff with corresponding rights and privileges

1.4 Expected benefits

- Why build your new system?
- How will the users benefit?
- What new knowledge is created?

(Sample) Initial analysis and user interview lead to the following expected benefits for the users

- The staff will have a tool to store and retrieve data and information they require to perform their jobs.
- The software system should lead to more efficient services to students and others, resulting in increase performance.

**Chapter 2 Feasibility Study**

2.1 Introduction

2.2 Problem statement

- Focus on specific problems and limitations of the current system
  - People, process, procedure, administration, etc.
- Fishbone diagram

(Sample) Currently the Student Services Division store data and information using paper forms and simple spreadsheets. This makes it difficult to access and retrieve information in a timely manner and also results in data isolation and inconsistencies. Students and others who interact with the SSD have to wait longer than it should be. When the staff who has been handling certain information is not available others do not know how to retrieve or access that information....

2.3 Related research and projects

- Summaries of research/projects that are related to your project
  - Specify how your project will be different from others
  - What restrictions/limitations/problems exist in previous systems?
  - Keep these documents for references

(Sample) There have been several projects related to this area including the Dean's Office Management System, the Scholarship Management System and CRM systems.

- The Dean's Office Management System was developed in 2000 by Mr. John Doe and Ms. Jane Flower. In this system, a small DB application was built using VB and MS Access. The application has the following functions....
- The Scholarship Management System by Mr. ABC and Mr. XYZ uses a Web interface with PHP as the programming language. MySQL was used as the DBMS. It provided features to store student family background and financial information. The system also allowed for loan payment history and

The Student Services Management System will use a Web client to access a DB server. To develop such a system, we will use interviews and questionnaires to collect user requirements and apply UML to model our system. Once our system model is finalized we will use ER modeling techniques and tools to design and normalize our database tables. The user interface will be designed to facilitate usability and to be attractive. This project will be different from previous projects in the following areas:

2.4 Requirement specifications

- You may have to go back and modify some of the scope that you submitted in the project proposal
- Give more details (fine tuning) of your scope
- More/change functions, features, supported platform(s), restrictions and limitations of your system
- infrastructure, related hardware/software requirements
- performance limitations, quantify if possible
- - Who are the users? Admin, staff, end users? How many? Concurrent?

(Sample) From initial evaluation of this project, our implementation will aim to satisfy the following scope

- Add, edit, delete data and fields related to student activities in the following areas
- extracurricular activities
- academic activities that is not handled by the registrar
- The user interface is web-based and can be used in both IE and Netscape browsers (version XX) and the HCI will be based on W3C standards
- The system will be tested on the following platforms: Windows 2000/XP, Red Hat Linux 7.x, ...
- PHP script version XX will be implemented to be compatible with XX Web server version ..
- Satisfactory speed performance will be based on running the server and clients in a 100 Mbps LAN environment
- The following reports ...

2.5 Implementation Techniques

- Specify the tools and techniques you will use in your implementation. You may need to change/add to your proposal/Chapter 1
- Programming languages, algorithms, tools
- Project management tools and documents, mile stones
- Identify areas you will need to study/practice further

(Sample) As can be seen from the Scope above, this project will utilize several Web and database technologies. Our implementation will attempt to use existing open source code and algorithms. In particular, the following tools and techniques will be used in this project

- PHP scripts version XX and JavaScripts version XX
- HTML 4.1 and XHTML
- MySQL for Linux RedHat version XX
- SQL queries will be used to process and retrieve data and information from the DB server as much as possible, this may limit Thai language support
- Table relationships will be checked and enforced though the DBMS

2.6 Implementation Plan

- Detail estimation of the time for implementing/performing each task (manhour)
  - Include the time needed to further study new areas/techniques

- PERT diagram, based on your time estimation
    - Consider the time needed for other activities (school, vacation, dates, family, etc.)

**Chapter 3 System Analysis and Design**

3.1 Analysis of the existing system

- Who does what? Include people and systems
- Work flow, architectural model, system flow
- This is written for comparing to your new system

3.2 User requirement analysis

- What do users want?
- How did you obtain this information? Survey, observation, questionnaires, interview, etc.

3.3 New System Design

- Various design documents: DFD, UML, new work flow
- State diagram, message flow diagram
- IPOS
    - User interface: this should covers concepts/format, NOT screen dumps
    - flow charts: for main steps or for complicated steps
        - may also be used for important algorithms
    - reports: formats and information to be contained in reports. This should include a list and the intent of each report. The actual report should be in the Appendix.
    - Data structures, ERD, DD
    - Site map: for web-related projects

**Chapter 4 System Functionality**

4.1 Introduction

4.2 System Architecture

- groups of system functions
- Modules and their functionality

4.3 Test Plan

- How to test each function? Commands, steps, scenario
- This should also include a sample of test scripts/forms

4.4 Test results

- To be added after (most of) the project is done and tested. This should be testing done in the testing phase, NOT testing in the implementation phase.
- Conclusions: which modules pass/fail, comments

**Chapter 5 Summary and Suggestions**

5.1 Introduction

5.2 Project Summary

- A summary of the products/results of your project
- What did you build? Did all of it work? Which features/functions are incomplete or missing?
- What performance goals did you achieve/fail?

5.3 Problems encountered and solutions

- Summaries of the problems that you encountered: time, skills, resources, project management, team work, etc.
- Were you able to solve the problems? How? Why not?
- Suggestions for preventing or solving similar problems.
- How can you improve your project management?

5.4 Suggestions for further development

- What can be improved about your project?
- What functions/features can be improved/added?

**References**

- A list of documents that you studied or referenced in implementing your project
- Web sites
- Use standard citation/reference style (see Thesis Preparation Manual)

**Appendix**

- Questionnaires
- Interview Scripts
- Test Scripts, Test Results
- Reports, Manuals, Etc.